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No. 135.—VOL. III.

SATURDAY, JULY 31, 1886.

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But if, like humble children, we would benefit by the goddess to the full, there are other things to attend to besides food and drink to make us comfortable in the dog-days. We are nearly all astray in the kind of raiment we wear, both in weight and colour and quality and substance. We draw down the divine caloric by dark, heat-producing clothes in a way which shocks Hygeia. Why not take to nankeen and cotton, and please the dear

soul, and comfort ourselves as well? She never wore funereal black in hot summers. She never had a hot chimney-pot on her head; she was never seen in ebony coloured trousers or a villainous hot mantle. She believes in white apparel, as angels ought to do—white window blinds and knickerbockers, white wide-awakes and sun-shades, white fish, white bread, white pulpy fruit, or as near that colour as possible, and white curtains and covers.

And Hygeia is right. Why should we keep such big fires and jets of lambent gas in the dog-days, consuming the life-giving oxygen, and yet complain of being overheated? He would be a plucky man who dare ride through public streets with white unmentionables, coat and vest, and white umbrella, on a white horse. He would look cool, however, and feel so; and if we could prevail upon ourselves to be a little lighter and whiter on saddle, or rail or steamboat, Hygeia says we should derive great joy thereby in July and August. At all events, we might make some approach to it in our dishabille. We need not be mere blocks for tailors and milliners to hang dresses upon, obliging us to be tight and uncomfortable because Fashion wills it. We require loose, lightly fitting garments, if we would keep cool.

Moreover, now that we are hobnobbing with goddesses and know their ways and philosophies, let us inquire why we open our windows and let in the broiling summer heat; and having let it in, why we do not allow it to go out again by the chimney or the roof. Limp, flabby girls, familiar to us all through Du Maurier's pencil, spend much time in stuffing our grates with lilies and peacocks' feathers and sunflowers. They fill the chimney with sacking and make the outside very pretty; thus no air makes its exit by the chimney flue. Hygeia says the young ladies are all wrong; and she doesn't care a fig for sunflowers, if they prevent the operations of nature. Hot air should ascend, and cool air come into a house. 'Dear girls,' says Hygeia, 'let these fads alone; pull out all the stuffing, and be natural. You are hot; cool yourselves. Why do you cram chimneys with flowers? It is not a festival. Make room for the king—for air, light, and comfort. Perish the peacock plumes; down with the gaudy flowers; and away with the fernery in front of them! Out with the sooty sacking. Give air, and plenty of it, in the dog-days.'

Hygeia says we don't make ourselves comfortable by the windows. We ought to have more green and white sun-blinds. We open our morning sashes and let in the bright heat all day, to make our bedrooms unbearable at

night. Nevertheless, everybody does it. Cottagers in the country open their lattices amongst honeysuckles, roses, and stocks; palaces are open amongst vines and trellises of wisteria and orange. Never mind, says our authority. Let me teach you to close all windows as closely as if they were glued, and let them remain so till the sun begins to wester in the heavens. We might do much by way of cooling our houses, if we attended to such sensible arrangements as closing in a southerly aspect, and opening in a northern one, always opening opposite the sun, and also by having free ventilation through the attics.

Directly the sun begins to decline, let every maiden and housewife, and man and woman and child, with an eye for the picturesque, and a feeling for health and beauty, throw up the Venetian or Parisian blinds. Open your rooms to the glories of the evening; throw up, and pull down the sashes; open wide all your doors. Let cool breezes enter into corridor and cellar and garret and room; let the 'caller' air circulate through every inch of the house hour after hour, whilst you are getting your evening meal, whilst you say your prayers, whilst you think of others after the toils of the day. If it be your priceless lot to dwell apart from city life, and have outside your cottage or villa or mansion, flowers, those lovely gifts of Dame Nature, let scents of rose and thyme come in at every gap in the hedge, at every rift of the wall, at every cranny of the house—scents of rosemary and mignonette, and lavender and bergamot, and lily and elderberry. Welcome the delicate perfume on its cooling, refreshing, healthy mission. It is Hygeia's gift—a superlative boon for the dog-days.

Strawberries are waiting to be plucked in all the hot months. If we have the possibility of enjoying a holiday, what can be better than a strawberry garden and plenty of cream? whilst larks aloft, and cuckoos in the shade, are singing in the plenitude of their full hearts, and whilst nimble fingers are spreading the white tablecloth on the grass to receive the dainty fruit.

Talk of lotos-eaters—we prefer strawberry gatherers. An old divine said he believed the joys of paradise would consist of eating strawberries to the sounds of a trumpet. We rejoice to think that we can have this transcendental pastime nearer home. We have the strawberries in full force, and there is generally a brass band round the corner to supply, for a small gratuity, the trumpet. Unluckily, doctors have decided that many of us derive no advantage from the strawberry; and alas! and alack-a-day! even

claret-cup and champagne and iced cream are occasionally proscribed! When boys, we ate more ices than we could afford; in maturity, we have the pocket-money—without the digestion. A lady in France thought that if strawberry ices were only *sinful*, no pleasure could exceed that which is to be enjoyed in the consumption of the pleasant fruit. In the eyes of some people, eating strawberries *has* become almost sinful, so the French lady will be able to satisfy her conscience, perhaps, on that score. Nevertheless, the old parson that Izaak Walton speaks of was right: ‘Certainly, God might have made a better berry than the strawberry, but certainly, God never did.’ So let us enjoy this heaven-sent fruit in the dog-days.

Not that we are at a loss for juicy fruits as long as we have our pine-apples and melons and tomatoes, our peaches and jargonelles, grapes and nectarines, and plums and apricots, a very paradisiacal melange, born of our glorious summer; all which indicates that providence nurtured them for the dog-days that we may eat and be satisfied. We may be sure that the sugar in fruits is modified by other elements, wisely elaborated by a Beneficent chemistry.

After the dog-days comes ‘St Luke’s little summer,’ beginning on the 18th of October and lasting for an octave. Horses and cows feel the heat, dogs whine, and cats show distress, birds sip the morning dew on the leaves for refreshment, even our trees and flowers hang their branches languidly. The Italians twit us by saying that only dogs and Englishmen walk in the sun. Well, it is so little of it we get, that we may be excused if we make the best of it, although we know we may suffer for our imprudence, and go home with colds or neuralgia from too free exposure and rapid cooling. Young dancing and gamboling Sylphs and Cupids in gauze, like so many butterflies in the sunbeams, had better be aware that they may get too much of it, although not often, and we must have an administrative check upon them, so that they do not fly into the heat and scorch their wings.

We are not an eminently sunny people; our fruit has not the rich orange tints of sunnier climes, where warmth is perennial and perpetual; and then dog-days come at last, and we go out to bask like lizards amongst the sand of our shores, or to splash amongst salt water at our bathing resorts. Our hot days ought to be an enjoyment, which they would be if we prepared ourselves for them and attended to the changes of temperature. We are not to throw off all our wraps in one grand effort to be free, still less to court

chills by foolishly hanging about damp places merely to get cool, and losing our animal heat quicker than we can replace it. Hygeia is the last person in the world to tolerate such errors. She requires us to use common-sense, and not to use an erroneous dietary; and if we obeyed her implicitly, our summers would leave us not so relaxed and overdone and dull and full of languor as they often do. If we will have heating food and heavy raiment, we resist the precepts of Hygeia, and we shall fail to win her smile when she draws the curtain for the season.

We must not tempt malaria by walking too late in dewy grass, when the moon is up, and all nature looks bright and beautiful, and only the nightingale sings or the willow-wren warbles amongst the osiers. We may stay out too late, by way of getting cool, until we get quite hot, and feverish with a cough that won't let us sleep; and as blackbirds and thrushes call upon us with dulcet notes about three o'clock A.M., we cannot answer the polite and musical invitation, if our throats suffer from the evening fog.

Young folk will pardon this dog-day talk, as it perhaps may benefit them. It is very pleasant to see them enjoying themselves, wild with the shimmering sunshine. We were all young once, 'before Decay's effacing fingers had swept the lines where beauty lingers,' and before rheumatism caught us in its horrible grip. Long may they enjoy themselves—and ourselves too enjoy our rollicking fun and nonsense amongst wild-birds and flowers and hayricks, amongst the scents of new-mown hay and clover and bean fields. What a lot of joy middle-aged people have to renounce, and yet we can still appreciate our dog-days!

An old proverb says, 'Every dog has his day;' but there are only forty dog-days in the calendar according to modern almanacs. They begin on the 3d of July, and end on the 11th of August. Bailey, the dictionary-maker of 1755, says the dog-days are 'certain days in July and August, commonly from the 24th of the former to the 28th of the latter, so called from the star Canis or Dog-star, which then rises and sets with the sun, and greatly increases the heat.' This was published three years after the introduction of new style, which took the place of old style in 1752. Another authority, more recent, says: 'The canicular or dog-days denote a certain number of days preceding and ensuing the heliacal rising of the Canicula or the Dog-star in the morning. Almanac-makers usually mark the beginning of the dog-days from about the end of July, and end them about their first week in September.'

Most people are accustomed to connect these days with mad dogs and hydrophobia generally, and they begin to think of M. Pasteur and his experiments at such times. There is evident confusion as to the time they begin and end. One thing is plain—they indicate the hotter portion of our year: some of them are so hot that we perspire if we stand still, though an Arab would freeze. What are we to do at such times? Simply, let us sit quietly if we can, and enjoy our siesta in a rather darkened room, with a pretty girl at the piano to sing for us, whilst we have our ‘hubble-bubble’ and rose-water or fragrant cigar and a pleasant book, till the cool of the evening. A considerable number of the dog-days are anything but hot; they are dashed by rain, as picnic parties know to their sorrow. St Swithin, of pluvial notoriety, bids us put up our umbrellas on the 15th of July, whilst he assuages the heat, and acts the part of Aquarius for the good of the world, spoiling all the custards and junkets and cheese-cakes, and taking out the stiffening of the ladies’ curls and collars in a remarkably disagreeable manner, by a sudden downpour, that often continues for many hours together. What an ungallant, heartless, and stingy old saint he must be!

IN ALL SHADES.

CHAPTER XXXVIII.

MR DUPUY was seated quietly at dinner in his own dining-room, with Nora at the opposite end of the table, and Uncle 'Zekiel, the butler, in red plush waistcoat as usual, standing solemnly behind his chair. Mr Dupuy was in excellent spirits, in spite of the little affair of the previous night, for the sugar-cane had cut very heavy, and the boiling was progressing in the most admirable manner. He sipped his glass of St Emilion (as imported) with the slow, easy air of a person at peace with himself and with all creation. The world at large seemed just that moment to suit him excellently. 'Nora, my dear,' he drawled out lazily, with the unctuous deliberateness of the full-blooded man well fed, 'this is a capital pine-apple certainly—a Ripley, I perceive; far superior in flavour, Ripleys, to the cheap common black sugar-pines: always insist upon getting Ripleys.—I think, if you please, I'll take another piece of that pine-apple.'

Nora cut him a good thick slice from the centre of the fruit—it is only in England that people commit the atrocity of cutting pine in thin layers—and laid down the knife with a stifled yawn upon the tall dessert dish. She was evidently bored—very deeply bored indeed. Orange Grove without Harry Noel began to seem a trifle dull; and it must be confessed that to live for months together with an old gentleman of Mr Dupuy's sluggish temperament was scarcely a lively mode of life for a pretty, volatile, laughter-loving girl of twenty, like little Nora. 'What's this, papa,' she asked languidly, just by way of keeping up the conversation, 'about the negroes here in Westmoreland being so dreadfully discontented? Somebody was telling me'—Nora prudently suppressed Marian Hawthorn's name, for fear of an explosion—'that there's a great deal of stir and ferment among the plantation hands. What are they bothering and worrying about now, I wonder?'

Mr Dupuy rolled the remainder of his glassful of claret on his discriminative palate, very reflectively, for half a minute or so, and then

answered in his most leisurely fashion: 'Lies, lies—a pack of lies, the whole lot of it, Nora. I know who you heard that from, though you won't tell me so. You heard it from some of your fine coloured friends there, over at Mulberry.—Now, don't deny it, for I won't believe you. When I say a thing, you know I mean it. You heard it, I say, from some of these wretched, disaffected coloured people. And there isn't a word of truth in the whole story—not a syllable—not a shadow—not a grain—not a penumbra. Absolute falsehood, the entire lot of it, got up by these designing radical coloured people, to serve their own private purposes. I assure you, Nora, there isn't in the whole world a finer, better paid, better fed, better treated, or more happy and contented peasantry than our own comfortable West Indian negroes. For my part, I can't conceive what on earth they've ever got to be discontented about.'

'But, papa, they *do* say there's a great chance of a regular rising.'

'Rising, my dear!—rising! Did you say a rising? Ho, ho! that's really too ridiculous! What, these niggers rise in revolt against the white people! Why, my dear child, they'd never dare to do it. A pack of cowardly, miserable, quaking and quavering nigger blackguards. Rise, indeed! I'd like to see them try it! O no; nothing of the sort. Somebody's been imposing on you. They're too afraid of us, my dear, ever to think of venturing upon a regular rising. Show me a nigger, I always say to anybody who talks that sort of nonsense to me, and I'll show you a coward, and a thief too, and a liar, and a vagabond.—'Zekiel, you rascal, pour me out another glass of claret, sir, this minute!'

Uncle 'Zekiel poured out the claret for his red-faced master with a countenance wholly unclouded by this violent denunciation of his own race; to say the truth, the old butler was too much accustomed to similar sentiments from Mr Dupuy's lips ever to notice particularly what his master was saying. He smiled and grinned, and showed his own white teeth good-humouredly as he laid down the claret jug, exactly as though Mr Dupuy had been ascribing to the African race in general, and to himself in particular, all the virtues and excellences ever observed in the most abstractly perfect human character.

'No,' Mr Dupuy went on dogmatically, 'they won't rise: a pack of mean-spirited, cowardly, ignorant vagabonds as ever were born, the niggers, the

whole lot of them. I never knew a nigger yet who had a single ounce of courage in him. You might walk over them, and trample them down in heavy riding-boots, and they wouldn't so much as dare to raise a finger against you. And besides, what have they got to rise for? Haven't they got everything they can ever expect to have? Haven't they got their freedom and their cottages? But they're always grumbling, always grumbling about something or other—a set of idle, lazy, discontented vagabonds as ever I set eyes on!’

‘I thought you said just now,’ Nora put in with a provoking smile, ‘they were the finest, happiest, and most contented peasantry to be found anywhere.’

There was nothing more annoying to Mr Dupuy than to have one of his frequent conversational inconsistencies ruthlessly brought home to him by his own daughter—the only person in the whole world who would ever have ventured upon taking such an unwarrantable liberty. So he laid down his glass of claret with a forced smile, and by way of changing the subject, said unconcernedly: ‘Bless my soul, what on earth can all that glare be over yonder? Upon my word, now I look at it, I fancy, Nora, it seems to come from the direction of the trash-houses.’

Uncle 'Zekiel, standing up behind his master's chair, and gazing outward, could see more easily over the dining-table, and out through the open doorway of the room, to the hillside beyond, where the glare came from. In a moment, he realised the full meaning of the unwonted blaze, and cried out sharply, in his shrill old tones: ‘O sah, O sah! de naygurs hab risen, an' dem burnin' de trash-houses, dem burnin' de trash-houses!’

Mr Dupuy, aghast with righteous anger and astonishment, could hardly believe his own ears at this unparalleled piece of nigger impertinence coming from so old a servant as Uncle 'Zekiel. He turned round upon his trusty butler slowly and solemnly, chair and all, and with his two hands planted firmly on his capacious knees, he said in his most awful voice: ‘Zekiel, I'm quite at a loss to understand what you can mean by such conduct. Didn't you hear me distinctly say to Miss Nora this very minute that the niggers don't rise, won't rise, can't rise, and never have risen? How dare you, sir, how dare you contradict me to my very face in this disgraceful, unaccountable manner?’

But Uncle 'Zekiel, quite convinced in his own mind of the correctness of his own hasty inference, could only repeat, more and more energetically every minute: 'It de trut' I tellin' you, sah; it de trut' I tellin' you. Naygur hab risen, runnin' an' shoutin', kickin' fire about, an' burnin' de trash-houses!'

Mr Dupuy rose from the table, pale but incredulous. Nora jumped up, white and terrified, but with a mute look of horror-struck appeal to Uncle 'Zekiel. 'Doan't you be afraid, missy,' the old man whispered to her in a loud undertone; 'we fight all de naygur in all Trinidad before we let dem hurt a single hair ob your sweet, pretty, white, little head, dearie.'

At that moment, for the first time, a loud shout burst suddenly upon their astonished ears, a mingled tumultuous yell of 'Kill de buckra—kill de buckra!' broken by deep African guttural mumblings, and the crackling noise of the wild flames among the dry cane-refuse. It was the shout that the negroes raised as Delgado called them back from the untimely fire to their proper work of bloodshed and massacre.

In her speechless terror, Nora flung herself upon her father's arms, and gazed out upon the ever reddening glare beyond with unspeakable alarm.

Next minute, the cry from without rose again louder and louder: 'Buckra country for us! Kill de buckra! Colour for colour! Kill dem—kill dem!' And then, another deep negro voice, clearer and shriller far than all of them, broke the deathly stillness that succeeded for a second, with the perfectly audible and awful words: 'Follow me! I gwine to lead you to kill de Dupuys an' all de buckra!'

'Zekiel!' Mr Dupuy said, coming to himself, and taking down his walking-stick with that calm unshaken courage in which the white West Indian has never been found lacking in the hour of danger—'Zekiel, come with me! I must go out at once and quell these rioters.'

Nora gazed at him in blank dismay. 'Papa, papa!' she cried breathlessly, 'you're not going out to them just with your stick, are you? You're not going out alone to all these wretches without even so much as a gun or a pistol!'

'My dear,' Mr Dupuy answered, coolly and collectedly, disengaging himself from her arms not without some quiet natural tenderness, 'don't be alarmed.

You don't understand these people as well as I do. I'm a magistrate for the county: they'll respect my position. The moment I come near, they'll all disperse and grow as mild as babies.'

And even as he spoke, the confused shrieks of the women surged closer and closer upon their ears: 'Kill dem—kill dem! De liquor—de liquor!'

'Ah! I told you so,' Mr Dupuy murmured, half to himself, very complacently, with a deep breath. 'Only a foolish set of tipsy negresses, waking and rum-drinking, and kicking about firebrands.'

For another second, there was a slight pause again, while one might count twenty; and then the report of a pistol rang out clear and definite upon the startled air from the direction of the flaring trash-houses. It was Delgado's pistol, shooting down the tipsy recalcitrant.

'This means business!' Mr Dupuy ejaculated, raising his voice, with a sidelong glance at poor trembling Nora.—'Come along, 'Zekiel; come along all of you. We must go out at once and quiet them or disperse them.—Dick, Thomas, Emilius, Robert, Jo, Mark Antony! every one of you! come along with me, come along with me, and see to the trash-houses before these tipsy wretches have utterly destroyed them.'

(To be continued.)

BEES AND HONEY.

THE honey-bee has been an object of great interest from the very earliest ages; the most ancient historical records make frequent reference to it. 'A little balm and a little honey' formed part of the present which Jacob sent into Egypt to Joseph in the time of the great famine. The 'busy bee' figures also in Greek as well as Hebrew history. The little creature has given a name to many females of high degree. The Hebrew name of the bee (Deborah) was given to Rebecca's nurse, as also to that magnanimous prophetess whose courage and patriotism inspired the flagging zeal and waning energies of her dispirited countrymen. The Greek name of the bee (Melissa) was given to one of the daughters of Melissus, king of Crete. It was she who, with her sister Amalthæa, is fabled to have fed Jupiter with the milk of goats. She is said, also, to have first discovered the means of collecting honey from the stores of the bees, from which some ancient writers inferred that she not only bore the name, but that she was actually changed into a bee.

Another Greek story tells of a woman of Corinth, also bearing the name of Melissa, who, having been admitted to officiate in the festivals of Ceres, the goddess of agriculture, afterwards refused to initiate others, and was torn to pieces for her disobedience, a swarm of bees being made to rise from her body.

The old Greek name for the bee seems to have fallen into disuse in this country as a name given to females, though there can be no reason why its use should not be revived, for it is at least as melodious as the Hebrew name of the same significance, still applied to many a matron and maiden—a name which is expressive of honeyed sweetness, as also of unwearied energy and untiring industry.

Those who have had personal knowledge and experience of bee-culture will bear out the remark that bees are not particular as to the size or the position of the home in which they choose to dwell, so that it suffices for them to carry on with security their wonderful operations. In their wild state, cavities of rocks and hollow trees are alike available; and in their domestic

conditions they have no preference for a straw skep over a wooden box, nor for the wooden house over the straw castle.

The bee, which, while under proper control and management, is one of man's best friends, proves, when assailed by him in any way, a terrible adversary. Allusion is made to this by Moses in his story of what befell the Israelites in their wilderness sojourn: 'The Amorites came out against you, and chased you as bees do, and destroyed you.' The strength and force of their sting is such as to enable them to pierce the skin of the horse and other large animals and kill them. Their ordinary speed when in flight, is from sixty to eighty miles an hour, and they have been known to fly past the windows of an express train when travelling at full speed in the same direction. Their manner of attack is to dash straight at the object aimed at; and commonly, when excited by the presence of some unknown spectator, and especially by the intermeddling of some undexterous or mischievous person, they will attack the face, aiming especially at the eyes. When, therefore, the thousands which inhabit a single hive are aroused by the sound of alarm, well understood by all the inmates, to repel an invader, they sally forth with a courage and determination which none can withstand, attacking their foes on every side with a fury it is impossible to resist. King David must have witnessed just such a scene, which he reproduces in his description of the fierce attacks, the determined onslaughts of his bitter and unrelenting foes: 'All nations compassed me about ... they compassed me about like bees.'

Somewhat recently, the mishap of a porter in handling a box of bees in transit by railway created an amusing and rather alarming scene at the station. There was a general stampede of passengers and officials flying in every direction, chased by the infuriated bees. It was only when some one, skilled in the management of bees, catching the queen and placing her in the box, restored confidence and quiet, for, flocking loyally to her standard, the whole colony returned to the case, which was in due time forwarded to its destination. But even this was a small affair compared with what is related in ancient history of persons being driven from their habitations, and the inhabitants of an entire town being compelled to flee before myriads of bees. Ælianus, who flourished about 200 A.D., gives an instance of this in one of his seventeen books on animals. Mungo Park, too, the African traveller, mentions a modern instance which took place near Dooproo: 'We

had no sooner unloaded the asses than some of the people, being in search of honey, inopportunately disturbed a large swarm of bees. They came out in immense numbers, and attacked men and beasts at the same time. Luckily, most of the asses were loose, and galloped up the valley; but the horses and people were very much stung, and obliged to scamper off in all directions. In fact, for half an hour the bees seemed to have put an end to our journey. In the evening, when they became less troublesome and we could venture to collect our cattle, we found many of them much stung and swelled about the head. Three asses were missing; one died in the evening, and another next morning. Our guide lost his horse, and many of the people were much stung about the head and face.'

The fierceness and unrelenting cruelty of the ancient Assyrians, and the terror with which their swarming multitudes filled the inhabitants of the lands they invaded, have caused them to be likened to bees in their much-dreaded attacks on such as have aroused their anger: 'And it shall come to pass in that day that the Lord shall *hiss* for the fly that is in the uttermost part of the rivers of Egypt, and for the bee that is in the land of Assyria. And they shall come, and shall rest all of them in the desolate valleys, and in the holes of the rocks, and upon all thorns, and upon all bushes.' The 'hiss' was simply a call, in allusion to the note of the queen bee, as she issues her royal mandate to her ever loyal subjects to prepare for action. It has also been supposed to allude to a custom prevailing in very ancient times in connection with bee-culture, or honey-raising in the neighbourhood of rivers. During the dry season, a number of hives would be placed on a flat-bottomed boat, in the charge of an attendant. Very early in the morning the boat would begin the day's voyage, gently gliding down the river, the bees sallying forth with the sun to collect their golden stores and deposit them in their several hives, which they commonly know by some mark. The innumerable flowers on the banks of the rivers offered them a fine harvest-field. At the approach of evening, the well-known whistle or 'hiss' of the care-taker—a decent imitation of the queen's own call—would bring them back to their hives in multitudes, when the boat would be paddled back to the farm or other place of rendezvous.

As an article of food, and as a much-valued and even royal luxury, honey has been used from the remotest ages. Nor was it much, if any, less in request as a healing medicine for both inward and outward application. And

though it may have fallen somewhat into disuse in these days, when many good things are overlooked, and when the artificial too often supplants the real, it may be safely predicted that the wide and rapid spread of bee-culture will induce a return to some of the wiser uses and methods and forms of adaptation employed by our early forefathers, as well as stimulate to new applications and new developments of its wondrous powers.

When and by whom mead or metheglin was first made from honey, could not be easily determined. The two words are not unfrequently applied to the same liquor; but that is not correct, as they are dissimilar. Both, however, are made from honey, sometimes also from the refuse or washings of the comb. Queen Elizabeth had such fondness for metheglin as to prescribe carefully how it should be made and with what a variety of herbs it should be flavoured. In Wales, it long continued to be held in high esteem; and its various beneficial properties have been quaintly set forth in a letter addressed to Cliffe the historian by the learned Welshman, Rev. James Howells (born 1594), brother of Thomas Howells, some time Bishop of Gloucester and Bristol. The uniqueness of the communication is the apology for its quotation in full:

SIR—To inaugurate a new and jovial new year unto you, I send you a morning's draught [namely, a bottle of metheglin]. Neither Sir John Barleycorn nor Bacchus hath anything to do with it; but it is the pure juice of the bee, the laborious bee, and king of insects. The Druids and old British bards were wont to take a carouse hereof before they entered into their speculations; and if you do so when your fancy labours with anything, it will do you no hurt; and I know your fancy to be very good. But this drink always carries a kind of state with it, for it must be attended with a brown toast; nor will it admit of but one good draught, and that in the morning; if more, it will keep a-humming in the head, and so speak much of the house it came from, I mean the hive, as I gave a caution elsewhere; and because the bottle might make more haste, have made it go upon these (poetic) feet:

J. H. T. C. Salutem et Annum Platonicum.

The juice of bees, not Bacchus, here behold,
Which British bards were wont to quaff of old;
The berries of the grape with furies swell,
But in the honeycomb the graces dwell.

This alludes to a saying which the Turks have, that there lurks a devil in every berry of the vine. So I wish you cordially as to me an auspicious and joyful new year, because you know I am, &c.

Metheglin is no doubt a healthy beverage, containing an admixture of milk. Pallus Romulus, when he was a hundred years old, told Julius Cæsar that he had preserved the vigour of his mind and body by taking metheglin inwardly, and using oil outwardly. Metheglin and mead may be made very strong, and, of course, they both contain some amount of alcohol. In Virgil's days, metheglin was used to qualify wine when harsh. He writes of

Huge heavy honeycombs, of golden juice,
Not only sweet, but pure, and fit for use;
To allay the strength and hardness of the wine,
And with old Bacchus new metheglin join.

Mead or metheglin was the nectar of the Scandinavian nations, which they expected to drink in heaven, using the skulls of their enemies as goblets. Thus we read in Penrose's *Carousal of Odin*:

Fill the honeyed beverage high;
Fill the skulls, 'tis Odin's cry!
Heard ye not the powerful call,
Thundering through the vaulted hall?
Fill the meathe, and spread the board,
Vassals of the grisly lord!—
The feast begins, the skull goes round,
Laughter shouts—the shouts resound.

In England at the present time, mead, like many other old and excellent domestic compounds, has passed almost entirely out of use. In very few houses could it now be found. Here and there in a farmhouse where old customs linger, it may still be had; and it is still used for colds and other complaints, both in the case of men and cattle.

The revival of bee-keeping and the conduct of the enterprise on scientific principles, will restore honey to its wonted place in the domestic economy; and if carefully studied and thriftily managed, the cultivation of bees and the product of honey may be made to form not only an important article of food and a considerable item of domestic revenue, but an ample source of amusement, and a means of recreation healthful alike to body and mind.

A GALLANT RESCUE.

SOME six years ago I was staying in a little village about half a mile from the sea, on the south-east coast of Cornwall. I had just recovered from a severe attack of blood-poisoning, and had not yet entirely regained my strength. My two companions were Herbert B——, a medical student, and Sam W——, a midshipman in the royal navy, both of whom had lived the best part of their lives at the seaside, and had been accustomed from their boyhood to boating and yachting in all sorts of weather. The former, about six feet in height, was a paragon of herculean strength. The latter, four inches less, was slightly but firmly built, and in his eyes there was a look of boldness and audacity which was unmistakable, whilst his every action gave evidence of a catlike activity.

That part of the Cornish coast on which we were staying was bare and rocky; a long line of cliffs rearing themselves straight out of the water to a height of about two hundred feet, stretched half-a-dozen miles on either side of us, affording no shelter for boats of a large size. The only thing resembling a haven was a small bay about a mile from our cottage, running a hundred and fifty yards inland, and facing south-south-east. From each side of this bay a bold reef of rocks jutted straight out to sea for about seventy yards, acting as natural breakwaters, and preventing a surf in the bay even in the roughest weather. In this bay, which was very dangerous of approach to those who did not know the landmarks, we kept a fishing-boat, about twenty feet long by six feet and a half beam; long and somewhat narrow, being lightly built, and meant for rowing as well as sailing.

I was sitting alone in the dining-room of our cottage about eleven o'clock on the morning of October 25, 1879. The wind, which had been blowing fresh for the past three days, had increased during the night to a strong gale from the south-west, and my two friends had gone out about an hour before to watch the very rough sea, and to see if there were any ships or boats in distress. I felt rather unwell, and was congratulating myself on not having gone out in such weather, when I heard a quick step outside the door, and Herbert burst in, crying in a decided manner: 'There's a dismasted schooner

drifting up channel, broadside on to the sea; there's a heavy squall of rain over Looe [the nearest port, about eight miles off], and the life-boat people can't see her; so Sam and I are going off to her in the fishing-boat; and as none of the villagers will come to steer, I've come to fetch you.'

'Fetch me!' I ejaculated, horror-struck. 'But my illness'——

'Put your illness in your pocket, and keep it there till you come back,' said my friend. 'You *must* come—unless you're afraid,' he added, glaring at me.

Although of a weak and nervous temperament, I am by no means a coward; so I told him I was ready to accompany him. On our way to the bay, Herbert told me that when first seen, the schooner was dismasted, but that the crew had managed to keep steerage-way on her by hoisting the jib and letting her run before the gale: the canvas being rotten, however, as is often the case on board small traders, the sail had blown right out of the bolt-ropes, and the vessel had swung round broadside on to the sea.

On reaching the cliff, a thrilling sight met my gaze. Some four miles off, a square-topsail schooner of not more than two hundred tons was being tossed about at the mercy of the waves. Her mainmast had gone by the board, and her fore-topmast had snapped off a few feet above the cap; her foreyard, however, still remained. She had a tremendous list to port—which was also her lee-side—and every sea that struck her broke clean over her, and seemed to shake her fearfully. We did not stop half a minute to observe this, but hurried to the bay where our boat was beached. Sam was preparing her for sea with all speed, but as coolly as if he were going out with a water-party on the upper reaches of the Thames.

After taking out some of the ballast to lighten her for the heavy pull—we could not sail, for wind and sea were dead against us—the boat was launched. No sooner had we got beyond the points of the two natural breakwaters, than a sea with what sailors call a 'head' on it struck us on the starboard bow, sending the boat's head flying round and filling her quarter-full of water.

'Gracious powers!' I cried, 'we'll never get out there. And if we do, we'll never get back safely with the boat full of people.'

'Pull her head round to the sea, Sam, my boy.—Mind your helm, Arthur, and don't talk,' said Herbert calmly. 'And as soon as we get beyond the

rocks, you can start baling,' he added, as we again met the first wave outside the bay. But this time I was prepared, and grasping the helm firmly, kept the boat's head dead on to the sea. With one vigorous stroke of the oars, which Herbert and Sam handled in a masterly style, we dashed over, almost through, the huge billow that threatened to engulf us, and not a moment too soon, for a second after it passed under our stern, it broke with a roar like the report of a cannon.

Then began a tremendous battle against wind and sea; Herbert dragging his oar through the water with that apparent ease and grace peculiar to men endowed with enormous muscular power; whilst Sam, who was pulling bow-oar, strained his sinewy arms and lithe body till, by their united efforts, the spray flew over the boat's bow as she boldly dashed over, often through, the waves. We were wet to the skin; and it was with great difficulty that I could keep the boat's head straight.

After about an hour and a half of as hard work as two men ever endured in a good cause, during which time I was kept constantly baling, we got close under the lee of the wrecked vessel, which had now drifted to within two miles of the shore. There were eight poor half-frozen wretches on board, one of whom was a woman, clinging to some spars which were securely lashed on the mainhold hatch. When we shouted and signalled them to throw us a rope, none of them moved. The cold and wet, and staying so long in the same position, had so stiffened them that they were unable to render us any assistance in getting on board. We then tried to approach the lee-quarter of the wreck; but just as we got under the mainchains, by which my companions meant to climb on board, a tremendous sea broke over the weather-quarter, and washing down over the lee gunwale, half filled our boat, and almost upset it.

'We're gone this time!' I exclaimed.

'Then we'll all go together,' cried Sam in a tone as if he rather enjoyed the idea than otherwise.

'Out oars and pull back again,' said Herbert calmly, without taking any notice of my frightened exclamation, for the wave had washed us some distance from the schooner.

On again approaching the wreck, we found the upper part of the fore-topmast floating about thirty feet from her side, with the fore-topmast backstay still fastened to it. After some trouble, Sam managed to cut the spar adrift and make the rope secure to our boat, the other end still being fast to the schooner. Herbert, telling me to keep the boat as clear of water as possible with the baling bucket, went forward. Taking hold of the rope, he jumped overboard, quickly drew himself hand over hand to the schooner's side, and climbed on board by the forechains. Sam soon followed him, though he was nearly washed away by a sea which broke over the schooner. Herbert, however, who was clinging to the foreshrouds, quickly grasped his wrist, and saved him.

After a short consultation, Sam went aloft with a rope, and lying out on the lee foreyard arm, passed the end of the rope through the brace-block. He then came down on deck again, and making a bowline on a bight (a knot with two large loops) with it, gave it to Herbert, who made it and the other end of the rope fast to a belaying pin. Sam then came back to the boat to help me to receive the unfortunates. Herbert proceeded with great difficulty to the main hatch, and waiting till a huge wave had washed over the schooner, took the woman in his strong arms and brought her to where he had made fast the tackle. He then signalled us to haul the boat as near to the wreck as we dared. Then he put the woman's head and shoulders through one loop, and her limbs through the other, and waiting his opportunity, swung her on to the boat, where we unslung her, so to speak, and passed the knot back to Herbert. The crew followed in the same manner. As Herbert was carrying the last of them down to swing him over to the boat, the schooner shipped a tremendous sea, which sent Herbert and his burden flying into the lee scuppers. After remaining in suspense for half a minute without either of them appearing above the bulwarks, Sam jumped at once overboard, dragged himself by the rope to the wreck, and climbed on board. Stooping, he disengaged the tightly clasped arms of the sailor from Herbert's neck; he then helped his friend, who was half insensible, to rise, and propped him against the bulwarks with his arms round the backstay. Sam was then about to stoop again to help the sailor, when he recoiled with an exclamation of horror. The poor fellow's head, as he had fallen with Herbert's huge weight on the top of him, had struck against the main-bits, and was shattered: he was stone-dead!

With great difficulty Sam managed to put his friend into the bowline and sling him over to the boat, he himself following by the rope by which we were made fast to the schooner. Herbert, who looked very pale and ill, sank back exhausted in the stern-sheets. A thin stream of blood was trickling from his temple; and he also suffered from pain in his right side.

It was late in the afternoon ere we cast off our rope and prepared for our homeward journey. We had scarcely got fifty yards from the schooners side, when a heavy sea struck her; she shook from stem to stern, then heeled over to port till we thought she would capsize; but she righted herself again, as if struggling to keep afloat, then slowly began to sink by the bow. A second wave struck her, more on the quarter; plunging her bow into the trough of the sea, she raised her stern in the air, and, diving like some sea monster, disappeared. We afterwards learned from the captain that her cargo—loose limestone blocks of about a hundredweight each—had shifted. The list this gave to the schooner had caused the mainmast, which was already slightly sprung, to go over the side, taking the fore-topmast with it. The shifting of the cargo had also started one of the planks, which accounted for the schooner springing a leak and going down.

The wind, which had chopped round to the southward, had blown us to within half a mile of the shore. Hoisting our close-reefed lug, we steered for the small haven, which we reached in safety in a quarter of an hour, after having narrowly escaped being upset by the ugly ‘topping’ of a wave at the entrance between the two points of rock. We were received with shouts of joy from the villagers and some coastguardsmen, who, having perceived that the vessel was drifting in shore, had prepared the rocket apparatus in case of emergency.

Poor Herbert had to be lifted out of the boat and carried to our cottage on a stretcher. A surgeon was in immediate attendance, and we awaited with no little anxiety the result of his examination. Three ribs were found to be fractured; but the wound in his temple proved very slight. Suffice it to say that our friend was able to return to his studies in a few weeks.

Neither Sam nor myself suffered from our exposure; the former remaining all night in attendance on Herbert; I, taking a steaming glass of grog, turned in between the blankets.

The shipwrecked crew were well attended to by the landlord of the village inn, and were next morning sent on to Plymouth. Nothing was known about the man who was killed; he had shipped on board the schooner at Falmouth, but no one knew where he came from.

A week after this event we received a letter of thanks from the owners of the schooner, who also offered us a handsome acknowledgment for our timely assistance, which we declined with thanks.

The captain, who is now master of a much larger vessel, and whose wife it was we had saved, insists on repeating his expressions of gratitude whenever we meet; but his tone becomes very grave when we laugh and attempt to make light of the danger we encountered.

OUR HEDGEHOGS.

WHO among us has not been amused and delighted by Frank Buckland's most original accounts of the various animals, wild and tame, with which, at different periods of his career, he came in contact? Reading in his *Life* the account of the hedgehog imported into the Deanery in the fond hope that it would devour the black beetles, has reminded us of some of our own experiences in connection with those animals. We were troubled with black beetles in our kitchen regions, and were informed that hedgehogs would eat them. It was long before the *Life of Frank Buckland* appeared; we had not the benefit of his experience, or we might have known that, as he says, 'they don't act. A hedgehog cannot possibly hold more than a pint of beetles at a time, and in my kitchen there are gallons of them.'

When the first hedgehog arrived and was turned loose in the kitchen, we expected great things of it; but, to our surprise, the creature would not take the trouble to catch the beetles. They might swarm on every side, 'beetles to right of him, beetles to left of him;' they might run right before his eyes—he only regarded them with placid indifference. He may have performed prodigies of beetle-catching in the middle of the night when no one saw him; but so far as our observation went, the only way in which he could be induced to eat any was when they were caught for him—taken up in the fire-shovel and presented to him on that as on a dish. Certainly there was no perceptible diminution in the number of black beetles, and our regret was therefore the less when before long the hedgehog mysteriously disappeared. Perhaps the beetles ate him; perhaps he managed to slip out unobserved into the yard. At all events, no trace of him was ever discovered; not even his skeleton in the flue, as was the case with Frank Buckland's hedgehog.

After this, I don't suppose we expected much in the way of beetle-eating from his successor, known amongst us by the name of Hogatha; but she was less shy and more sociable than many hedgehogs, and amused us by her droll ways. She would of course roll herself into a prickly ball when touched, but would uncurl as I sat with her on my lap, and look about her with her bright little eyes. I think she would soon have become tame, and I

should have made a pet of her, but for one unfortunate circumstance. If even the whale has his unmentionable parasite, it will not perhaps appear surprising when I mention that fleas in great number inhabited my little friend's bristly coat. When she uncurled as she lay on my lap, they could be seen running in and out over her odd little head and face. Perhaps this is a favourite locality, being less bristly, and presumably more comfortable for the fleas than the more prickly portions of the body. But it was too much. Not even for the sake of cultivating the acquaintance of the charming Hogatha, could I face the prospect of restless nights and irritated skin, so our friendship waned.

It must have been this hedgehog which frightened me one night. I was not learned in natural history, and didn't know that hedgehogs could run fast and mount stairs. It was late at night, and I was in bed, when I was startled by hearing strange noises in the passage outside my door. Sometimes they appeared distant, sometimes near; sometimes there came a kind of scraping at the door, which had a most uncanny sound. Is there such a thing as being physically superstitious, the mind having little or nothing to do with it? If so, I was physically superstitious; and the tendency which was in my blood, handed down perhaps from old Breton ancestors, was developed (parents and nurses, please take heed to my words!) by ghost stories told me in my childhood. At the time of which I am writing, though quite grown up, I well remember there was one story in particular I hardly dared recall, which, if it came back to my memory in the night, would cause the old feeling of terror to overwhelm me like a flood; wherefore it was with an effort that I got up and lit the gas; then, 'taking my courage in both hands,' I opened the door—and behold! there was Hogatha tearing up and down the long passage like an express train! I couldn't consent to have her *and* her fleas, and I couldn't have her without, so I conveyed her down-stairs, and shut her in the kitchen.

Then there was the sweet little baby hedgehog, given me by a lad who found a nest in his garden. We didn't mean to be cruel, either of us, but no doubt were so, for the poor little thing was too young to be taken from its mother. I could not induce it to eat or drink, and at last I gave it to the cat, which had kittens at the time, to see if she would adopt it. She received it graciously when I put it into her basket, as though it had been her own kitten. But it was all no use; the poor little thing pined and died.

We were by this time pretty well convinced that beetle-eating on the part of hedgehogs was chiefly theoretical, with just as much relation to the realities of life as many other theories, and no more. We desired, however, to keep our minds open to new impressions; and when told that they were useful in a garden because they would eat the snails and slugs, we believed our informant, and hailed with gratitude the arrival of two fresh hedgehogs. They were named Paul and Virginia, and were shut up in the summer-house, with the idea that when they had become well accustomed to that as their place of abode, they might not run away when allowed to go loose in the garden. But there must be some mistake about their fondness for snails and slugs. I took one to Paul (or Virginia, I am not sure which) one day; and, after some hesitation, he slowly ate it; but presently threw it out of his mouth. It didn't seem encouraging when you remember that they were expected to help to clear the garden of such pests. However, Paul and Virginia were allowed, when supposed to be sufficiently at home, to take their walks abroad, and then they also disappeared, nor have I ever seen either of the queer creatures since.

HOW PAT DELANEY PAID HIS RENT.

I WAS born in County Blank, Ireland, educated in Dublin, and chose for profession—if profession it may be called—that of a tea-planter; but times were bad, health failed me, and after ten years spent in Assam, I returned to England with the intention of remaining, should a suitable appointment be procurable. No one knows, however, till he tries how difficult it is to find suitable employment on returning after a lapse of years to one's old haunts; the true reason of it being that there is too large a proportion of the *genus homo* collected together in this corner of the globe. My parents had died during my absence, and their property had passed into the hands of an elder brother with whom I was not on good terms, so I did not revisit the old place. Hearing, however, that my uncle, Sir Toby O'Bride, who owned considerable estates in another county, was having some trouble with his tenants, I thought I would cross over and see him.

My respected relative was in the act of shutting up house and beating a hasty retreat from the country. No rents had been forthcoming for some time, so he had lately changed his agent. The new one succeeded in bringing a few of the tenants to their senses and the rents to Sir Toby's pocket, but two nights previous to my arrival the unfortunate man was shot when returning home through the park, after dining with Sir Toby. The police had some *suspects* in charge; but as it proved, they had no hand in the affair, and the guilt was never brought home to the real perpetrator.

'I don't know,' said my uncle, 'what is to be done, but at present I intend going away for a time. They will shoot at me next, if I remain. This shocking affair has quite unnerved me.' My uncle did indeed look shaken and ill.

'I have a plan,' said I, 'if I may suggest it? Let me take the agent's place, and see if I can improve matters. The people all know me more or less, and if any of them try to make holes in me, they will find me well prepared to retaliate. I mean this seriously, uncle. I am an idle man at present, and will be more than pleased if you let me have my way.'

He pooh-poohed my proposition at first, declaring it was simply suicidal to attempt such a thing; but he finally consented, and I was installed in the agent's cosy cottage at a salary of four hundred pounds per annum. The first step was to purchase ostentatiously a pair of six-chambered revolvers, and erecting a target in the garden, I peppered away at it. Whenever any one came to my office, I took occasion to show what an excellent shot I was. The office window stood high from the ground, and was furnished with iron bars and a grating like that of a prison cell. When the tenantry came to pay their rent, they found me seated at the table with one of my faithful beauties on each side of me, and it was well known that I never left the house without them.

Whether it was owing to my knowledge of the character of the people with whom I had to deal, or whether it was their knowledge that I liked them sincerely, but knew them too well to be 'done' by them or to fear their threats, I cannot say; whichever way it was, no attempt was made on my life, and a larger proportion of the rents due passed through my hands in the course of the year than had through those of the agents for some time previously. Of course, there were some tenants who could not or would not pay their rent. Stories of bad harvests, cattle dying, pigs getting measles, and starving families at home, came eloquently from the glib tongues of the delinquents. Sometimes true, more or less, generally less, for there was very little bad land on the estate. Foremost amongst the last-named section was one named Delaney. He held a good farm, which had been tenanted for generations back by Delaneys, who had been counted good tenants in their day; but this Pat came under the influence of agitators, who perverted his ideas of honesty.

Pat Delaney was among the first to refuse to pay his rent, and the aggravating part of it was that I felt sure he had the money. He was the best judge of horses in the country-side, and attended all the fairs, doing a good deal of cattle-dealing in a quiet way, so that in spite of bad seasons, he was counted a well-to-do man among his fellows. But on rent-day not a shilling was forthcoming. The old story—failure of the potato crop, bad harvest, wife sick, a lot of mouths to fill, and 'Wouldn't I put in a word for him with the masher? Shure, the kind ould masher wouldn't be hard on a poor man. He would pay up next rent-day for sartain.'

‘No, Pat,’ said I. ‘This is the second time you have brought me that story. You are far behindhand with your rent; and if you don’t pay up now, out you must go. The land is good and the rent low. If you can’t make it pay, we must find another tenant who will. It goes against my heart to turn you out, for Delaneys have been on the place for three generations now; and I am sure you *can* pay, if you like. The Delaneys were never paupers before.’

Glancing sharply at him, I saw a flicker of indecision pass over his countenance, and his hand fidgeted with the edge of his jacket; but in a moment the former expression of doggedness came over his face like a cloud; he straightened himself, and said insolently: ‘Shure an’ wouldn’t I pay if I could? It isn’t dishonest ye’re thinkin’ I am?’

An idea struck me. Changing my tone, I remarked indifferently: ‘O no; the Delaneys were always honest. But if the money is not forthcoming, out you must go, and there’s an end of the matter.’

Gathering up the books, I returned them to the safe, locked it, and taking my hat, I turned to my companion and began confidentially: ‘I want to ask your opinion about something, Pat. They tell me you are a good judge of a nag; I want you to tell me what you think of one I have in the stable just now.’

At the word ‘nag,’ Pat was all attention.

‘She’s a beauty, and, I imagine, should fetch a good deal. She belongs to a friend of mine, who is hard up, and asked me if I could sell her for him, which will be easily done; but I want your opinion of her. There are two or three offers for her already. She was bought, I know, for one hundred and twenty pounds; but that is a little time ago; and my friend would take sixty pounds for her now, or even forty pounds, down.’

Pat’s eyes scintillated, and I saw his hand tremble with eagerness. By this time we had reached the stable where Black Bess, my beautiful hunter, stood. She had arrived a week before, a gift from my uncle, Sir Toby, and she looked her hundred and twenty guineas every inch, the beauty!

‘Cheap at forty pounds, eh, Pat? Look at her points, man. I wish I could buy her myself.’

‘She’s a purty crayture, sor,’ ejaculated Pat as he went over her points with keen appreciation. Looking at her teeth, patting the glossy, arched neck, and

finally passing his hand down each leg, he raised his head, and said in a sheepish sort of way: 'She's worth her forty pounds, sor.'

'Yes, I know that. Now, I thought you might know of some one wanting a horse. Perhaps one of your friends might like to deal; but I must have cash down.'

'I know ov one man who moight take him, sor.'

'Do you? Well, I'd be glad if you'd send him to me to-morrow; and if the mare is still here, he may have her; but he must take his chance, mind you. I have several offers, and "first come first served" is the rule for this business. Sir Thomas Clarke has an eye on her, and would probably give sixty pounds if I hung on a bit; but my friend wants the money at once. Emerson of Bogside was here this morning, and liked the looks of her; said he might look back in the afternoon and close the bargain; so your friend must take his chance.'

'Shure, sor, and ye moight jest keep her till me frind sees her to-morrow. He's sartain shure to take her, and cash down on the spot.' Pat was most persuasive, and I saw by the gleam in his eye that he was safe on my hook. He knew as well as I did that he had only to take her to the first fair and he would get seventy or eighty pounds for her, if not more.

'No, no. A bargain is a bargain. I told Emerson that it would be a case of first come first served. If Black Bess is here to-morrow, your friend can have her, and welcome; but I cannot keep her for any one.'

A heavy footstep tramped up the garden path, and we heard a loud voice asking for me.

'Why, that must be Emerson back already!—Good-day, Pat; I don't think I need ask you to trouble your friend, after all.'

'Stop, stop, sor; I'll buy the mare meself, and here's the money.' Ripping open the lining of his jacket, he thrust a roll of dirty notes into my hand.

Slowly I counted them, 'One, two, three, four, five, six, seven, eight five-pound notes. That makes forty. Thanks, Pat. Just half your rent! Now, you go home and bring me the other half. I know you have it all, and you cannot deny it.'

When I wrote to Sir Toby, I had the extreme satisfaction of telling him that Delaney had paid up in full; and Black Bess carries me none the worse for having been an unconscious actor in the little drama which proved so successful.

THE MONTH: SCIENCE AND ARTS.

WE have much pleasure in recording the establishment of the 'County Scientific Society for Middlesex.' There are many such Societies, most of them in a very flourishing condition, dotted about the kingdom, where, for a small subscription, the members can meet at lectures, concerts, and various entertainments. In addition to this, many of these institutions have attached to them educational and art classes, which students can attend for a small fee. It is certainly time that the metropolitan county should be similarly provided for, although for some years past many local institutions of the kind have sprung up round about the great city. Among the vice-presidents of the new Society we note such honoured names as Lubbock, Huxley, Flower, Abel, and Geikie. These alone should insure that success which we hope the enterprise will achieve. Application for membership and other particulars may be obtained from Mr Sydney T. Klein, Clarence Lodge, Willesden, N.W.

The newspapers constantly remind us that there are many persons in the kingdom who object to vaccination, and, as a matter of course, there are not wanting agitators who are constantly calling aloud for the repeal of the law which makes the operation compulsory. Three years ago an outcry of the same kind arose at Zurich in Switzerland, with the effect that the cantonal law of compulsory vaccination was repealed. By reference to the official returns set forth in a paper by Professor Dunant, we are able to judge of the effect of the popular vote. In the canton named, the deaths from smallpox were in the year 1881, seven; in the two following years there were no deaths from that disease; in 1884, they rose to eleven; in 1885, they were seventy-three; and in the first three months of this present year, the deaths from smallpox were no fewer than eighty-five. These terrible figures need no comment, save the remark, that they do not take into account the sightless eyes and dreadful disfigurements of those who were attacked but did not die.

More conclusive evidence as to the efficacy of Jenner's discovery may be gathered from Dr Jassen's book, recently published at Brussels. Let us quote one instance given. Last year, in twenty-one German towns having an aggregate population of four millions, where vaccination was compulsory, the deaths from smallpox numbered twenty-seven; while in fifteen French towns owning the same aggregate number of inhabitants, but where the law was not in force, there were no fewer than eight hundred and sixty-six deaths from smallpox in the same period.

According to a Report published by Lieutenant von Nimptsch of a journey made by him with a traveller attached to the Congo Free State, a navigable river has been discovered by them which is likely to be of great importance to the future trade of the Congo. The river Congo, as will be seen by the map, flows in a north-westerly direction, and afterwards takes a southward course to its mouth in the Atlantic Ocean. Within the large tract of country comprised in this bend of the river, has been found the new waterway. It is described by the travellers as flowing through wide plains well adapted for cultivation, with pasturage, and forests of palms, and gutta-percha trees. Plenty of ivory was obtainable, in exchange for empty boxes and tins, from the inhabitants of the many villages which lined the banks of the river. There are many affluents to this waterway, one of which was navigable for two hundred and fifty miles. Altogether, we have presented to us in the Report a network of navigable rivers extending over a length of more than three thousand miles.

An interesting note in the *Times* tells of a place in Russia, in the region of the Transbaïkal, where there exists a multitude of mineral springs. These have been held in high repute by the natives for many years, and it has long been the custom to bring patients to the springs for curative treatment. Not only human beings, but cattle, sheep, and horses suffering from cutaneous affections have, it is alleged, benefited by such treatment. The temperature of the springs varies from thirty-five to over a hundred degrees Fahrenheit; and some are ferruginous, some alkaline, and others sulphurous in composition. At present, the alleged virtues of these waters are only known locally, and there is little accommodation for strangers. But it is believed that, in the future, patients will be attracted to the place from great distances.

At Sonnblick, one of the heights of the Tyrolese Alps, the summit of which is ten thousand feet above the sea-level, an observatory is in course of construction, which will represent the highest establishment of the kind in Europe. The summit of this mountain is more easily accessible than some of the neighbouring peaks; and there is already a wire-rope way which affords communication with some mines half-way up the mountain. It was the owner of these mines who was the first to point out the desirability of establishing an observatory here. The building will consist of a blockhouse and a massive stone turret forty feet high, which will form the observatory proper. The house is being built of timber in preference to stone, as experience teaches that the former material is more effectual in keeping out the intense cold prevalent at such an altitude. The observer will be in telephonic communication with the miner's house two thousand feet below him; and from the latter place a record of his observations will be telegraphed to the nearest city, and thence all the world over.

Another portion of the old wall of London has recently been laid bare by some excavations now in progress near Ludgate Hill, at the Broadway, Blackfriars. This portion of the ancient defence of the capital is clearly a continuation of the fragment removed a few years ago, and is built mainly of limestone and rough mortar intermingled with tiles, bricks, and, strange to say, lumps of soft white chalk.

We have lately had the opportunity of examining a little piece of apparatus which represents the most recent advance in photographic contrivances. In outward appearance it is a book, somewhat less in size than the ordinary two-shilling railway novel. Upon opening it, it is seen to have flexible folds like the web of a duck's foot, and when open, it remains so fixed by invisible springs. It is in reality a wedge-shaped camera furnished with a lens, which is sunk into the middle of the back of the imitation book. It is also furnished with a hidden shutter, which closes and uncloses the lens aperture at the will of the owner.

The recent inclement and unseasonable weather in the south of England has been characterised by two very unusual occurrences. First, at Deal in Kent, a small whirlwind lifted some boats from the beach, displaced a heavy crane on the railway, and did other damage. A few days afterwards, a similar phenomenon occurred at Sparham, Norfolk, which presented some extraordinary features. Its course could be traced for half a mile; and its

path of destruction was well marked by a patch which, commencing with a width of two yards only, finished at the end of the half mile with a width of one hundred yards. During the two minutes which the storm lasted, it uprooted trees, unroofed houses, pulverised some hencoops, and wrought much destruction. The weather was perfectly calm except over the space covered by the whirlwind.

The total eclipse of the sun which will take place on the 29th of August is to be observed by an expedition sent out by the Royal Society and by funds from the Treasury. The party will at first proceed to Barbadoes, and will be conveyed thence to Grenada by a war-vessel. The island will be covered with stations for observing the eclipse, and all modern instruments will be used in the operations. The eclipse will not be visible at Greenwich.

There has been established for many years a school of practical engineering at the Crystal Palace, Sydenham, and this school has already educated many who have excelled in their profession. As an example of the practical method of instruction pursued, we note that recently a steamer of thirteen hundred tons was worked from London to Dundee and back by a division of the students who are turning their attention to marine engineering. They were divided into gangs of four, and each gang had to work for ten hours in the engine-room under the strictest discipline. While in the north, they had an opportunity of making a professional inspection of the new Tay Bridge.

Experiments have recently been made at Berlin with a new description of military shell which is charged with rolls of gun-cotton. The projectile is said to be so destructive that no defensive works however solid can withstand it. The German government are so satisfied with the experiments that they have ordered a large number of the shells to be manufactured forthwith.

According to the *Revue Scientifique*, the discovery or suggestion of the Germ theory of disease cannot be placed to the credit of modern physicists, but is due to a Dr Goiffon, who died at Lyons more than one hundred and fifty years ago. He published a work on the Origin of the Plague in 1721, from which the following is quoted: 'Minute insects or worms can alone explain these diseases. It is true they are not visible, but it does not therefore follow that they are non-existent. It is only that our microscopes are not at present powerful enough to show them. We can easily imagine the

existence of creatures which bear the same proportion to mites that mites bear to elephants. No other hypothesis can explain the facts; neither the malign influence of the stars, nor terrestrial exhalations, nor miasmata, nor atoms, whether biting or burning, acid or bitter, could regain their vitality once they had lost it. If, on the other hand, we admit the existence of minute living creatures, we understand how infection can be conveyed in a latent condition from one place, to break out afresh in another.'

Among the multifarious objects on view at that palace of wonders, the Colonial and Indian Exhibition, are naturally many products of the animal and vegetable kingdoms which are comparatively strange to British experience. Among these may be named certain drugs, gums and resins, oils, dyes, different kinds of timber fibres, leathers, &c. Now, it is evident that many of these things may be useful to our manufacturers if only their properties can be made known. With this view, arrangements have been made for the systematic examination of these foreign products, to see whether they can be applied to present manufactures, or whether they are suitable for new purposes. Visitors to the Exhibition can attend these examinations, which, if necessary, will be followed by conferences.

A thoughtful man, in strolling through the vast network of galleries at the Colonial Exhibition, cannot help feeling that there is some excuse for the national boast that 'Britannia rules the waves,' for all the treasures of the earth seem to be gathered together here. The next thought that must occur to every one is the regret that the Exhibition is only a temporary one, and that the riches which have been gathered with such care and trouble from such a wide area must soon be again dispersed. There are indications that this regret, felt as it is by the executive as well as by the casual visitor, may lead to a practical result. For years it has been urged by a few that London ought to possess a Colonial Museum. We have now an unusual opportunity for forming the nucleus of such an establishment, and that opportunity should not be lost.

It seems difficult to believe that in these hard-working and matter-of-fact times, persons should be found who revert to the gross superstitions common to the people in far-off centuries. A so-called astrologer has been for a year at least making a good living by casting nativities in the neighbourhood of Brunswick Square, London; but his operations have been cut short by a fine in the police court.

The controversy which has been going on for some months between Mr J. C. Robinson and Sir James D. Linton as to the alleged fading of water-colour paintings through exposure to light and other influences is now to be brought to public arbitration. Sir James Linton, the President of the Royal Institute of Painters, has arranged to open an exhibition of the works of the most celebrated artists of the last fifty years, so that all may judge whether they have deteriorated. He is a champion for the permanence of this delightful phase of art; while Mr Robinson thinks differently.

The manufacture of whitelead, while representing one of our most important industries, has always had the bad character of being most destructive to the health and lives of the workmen employed in it. The substitution of other materials in the making of white paint has been constantly tried, but all give the palm to whitelead because of its 'covering' power. A new process has just been devised by Messrs Lewis and Bartlett for producing whitelead of the finest quality direct from the ore. The process is too long to describe here, but we may briefly state its advantages over the old method. It combines two manufactures, for whitelead and piglead are produced simultaneously. No deleterious fumes escape into the atmosphere, for the smelting furnace employed has no chimney. The operations are conducted with a greatly reduced expenditure of time and labour; while, best of all, the industry is not in any way hurtful to the workers. The process is an American one, and is introduced into this country by Messrs John Hall & Sons of Bristol.

It would seem from the letter of a correspondent to the *Standard* that frogs and mice are deadly enemies. This gentleman observed a battle-royal going on between these creatures in a shed. The mice pursued the frogs all over the place, for some little time without result, for the frogs managed to elude them. But gradually the mice gained an advantage, capturing and recapturing the frogs, and biting them until they were incapable of further resistance. The mice then finished the business by devouring a portion of the dead frogs.

The last new agricultural implement is a hay-loader, which has been recently patented by Mr Spilman of Dakota. This machine collects the scattered hay from the field, raises it to a suitable height, and finally discharges it upon the hayrack of the wagon. Lovers of the beauties of the country will regret that the pleasant sight afforded by a number of bronzed

haymakers loading a wagon, a scene which has so often tempted the artist's pencil, should be threatened by the introduction of this mechanical thing. But time is money, and there is now little room for sentiment.

From the Report of a Cattle Show recently held at Buenos Ayres we learn that the South Americans are by no means behind Europeans in their use of machinery and implements for agricultural use. Also, that the live-stock there has much benefited by the importation of short-horns from Britain, and from Charolais in France, so far as the cattle are concerned, and that the sheep have equally benefited by acquaintance with our southdowns and with the French merinos. Some few years ago, a loud outcry arose among our agriculturists that buyers from the other side of the Atlantic were purchasing all our best stock at prices far beyond what the British farmer could afford to pay. There is now the hope that we shall be recouped by the importation of mutton and beef of first-rate quality. The freezing process has now been brought to such perfection that, with meat from the English stock, it should afford us the opportunity of getting the best flesh food far cheaper than we can attempt to raise it for ourselves.

Surely Mr Flinders Petrie is the most successful and energetic digger that the archæological world has ever seen. His past discoveries have already resulted in much increased knowledge of dead nations; but now he has lighted upon a most curious find in the north-eastern delta of the Nile: this is a royal palace, which is identified with the greatest certainty with that building which the Bible calls 'Pharaoh's house in Tahpanhes.' The building carries us back in imagination to the Egypt of two thousand five hundred years ago. Next to its scriptural connection, interest centres in the description of the domestic offices of the building; and as we read of the kitchen with its dresser, the butler's pantry full of empty wine vessels and their stoppers, the sanctum of the scullery-maid with its sink, we feel that the place has been tenanted by ordinary human beings. Mr Petrie's account of the sink is worth quoting: 'It is formed of a large jar with the bottom knocked out, and filled with broken potsherds placed on edge. The water ran through this, and thence into more broken pots below, placed one in another, all bottomless, going down to the clean sand some four or five feet below.'

Mr Francis Greene publishes in an American journal the results of some careful observations which he has made on street traffic. According to him,

asphalt is a far better covering for roads than either granite or wood. He puts the matter in this way: a horse will travel five hundred and eighty-three miles on asphalt before meeting with an accident, four hundred and thirteen on granite, and two hundred and seventy-two miles on wood. This agrees with experience in London with regard to the first two materials, but not with regard to wood, which experts say is the safest material of all. Londoners have certainly the best means of judging of this, for there is very little wood-paving in America. At the same time, it is quite certain that altogether accidents are far more frequent in London. This may be accounted for by the dampness of the air, which gives rise to the peculiar greasiness of the streets, so fatal to horses; and also by the increased traffic, which leads to the accumulation of manure, another element in the slippery state of the roads.

The snail harvest has recently begun in France. The 'poor man's oyster' is so appreciated by our neighbours that Paris alone consumes some forty-nine tons daily, the best kind coming from Grenoble or Burgundy. The finest specimens are carefully reared in an *escargotièrre*, or snail-park, such as the poor Capuchin monks planned in bygone days at Colmar and Weinbach, when they had no money to buy food, and so cultivated snails. But the majority are collected by the vine-dressers in the evening from the stone heaps where the snails have assembled to enjoy the dew. The creatures are then starved in a dark cellar for two months, and when they have closed up the aperture of their shell, are ready for cooking. According to the true Burgundy method, they are boiled in five or six waters, extracted from the shell, dressed with fresh butter and garlic, then replaced in the shell, covered with parsley and bread crumbs, and finally simmered in white wine.

OCCASIONAL NOTES.

THE PRESENT POSITION OF THE PETROLEUM INDUSTRY.

MR CHARLES MARVIN, one of the first to direct attention to the Russian petroleum fields at Baku, in speaking lately of the transference of petroleum in bulk, said that America was at present the principal petroleum power. By the development of the petroleum fields at Baku, Russia had recently sprung into the position of the second petroleum power; and Mr Marvin thought that England should come to the front and occupy the third position as soon as possible. By the annexation of King Theebaw's dominions, we had come into possession of the Burmese petroleum fields, and he thought steps ought to be taken at once by the Indian government to survey these fields and to throw them open to British capital and enterprise. Within the last few years, since the extension of the railway, considerable petroleum deposits had been discovered in Beluchistan, but he regretted that the Indian government had decided to make them a Crown monopoly. Still more recently, petroleum in abundance had been discovered in Egypt. Since he wrote in 1882 of the Caspian petroleum fields, eighty steamers had been placed on that inland sea to carry oil in tanks from Baku to the mouth of the Volga; and on the Volga there were upwards of a hundred vessels running. At present, nearly all the petroleum arriving in Europe from America was brought in barrels; several tank steamers were, however, being constructed on the Tyne for the purpose of carrying petroleum in bulk.

Mr Phillips, in lecturing at the Royal Aquarium on this subject, said that the total shipments of refined oil from America in 1885 amounted to 6,985,637 barrels, of which the United Kingdom received 1,269,723; London taking 666,964 barrels. If the total shipments were placed in barrels end to end, like a string of beads, they would reach from London to New York. It is estimated that the world's consumption of illuminating oil amounts to 1,800,000 gallons every day. At the present price of oil as sold retail, and taking an ordinary circular-wick burner of forty candle power, it costs about three-sixteenths of a penny per hour, which was fifty per cent. cheaper than

gas. In this connection, it may be mentioned that the Balloon Society of Great Britain is offering a prize for a cheap safety-lamp suitable for universal use. The annual production of mineral oil shale has continued to increase in Scotland, until in the present year it stands at the unprecedented figure of about two million tons.

THE CRYSTALLISATION OF FRUIT.

From a paper by Consul Mason, of Marseilles, we learn a good deal about the business of preserving fruits by the crystallising process peculiar to South-eastern France, and practised on a large scale at Apt in the department of Vaucluse, at Clermont in Auvergne, as well as at Marseilles, Grasse, Avignon, and other places. It is curious to find these preserved fruits exported not only to England and the United States, but also to other countries, such as Algiers, the East and West Indies, and even South America, where nature has made the dwellers so far independent of preserved fruit. The fruits preserved by the crystallised process are chiefly pears, cherries, apricots, pine-apples, plums, figs, citrons, oranges, melons, and a dwarf orange called 'chinois.' Peaches are found to be too costly to be treated to any extent in this fashion.

For the purposes of crystallisation, the fruit must be fresh, clear of all decay and blemish, and of the proper degree of ripeness. The chief thing to be done in this process is to extract the juice of the fruit and replace it in the pulp with liquid sugar, which, upon hardening, not only preserves the fruit from fermentation and decay, but retains it in its original form and consistency.

The fruit is first carefully assorted in respect to size and uniform degree of ripeness. Pears, pine-apples, and quinces are pared; citrons are cut into quarters and soaked a month in seawater; and the 'pips' of apricots, cherries, and peaches are carefully removed. This work, which requires a certain degree of skill, is chiefly done by women. When thus prepared, the fruit is immersed in boiling water, which quickly penetrates the pulp, dissolving and diluting the juice, which is thereby nearly eliminated; then the fruit is taken from the water and drained, leaving only the solid portion of the pulp intact. The period of immersion must be regulated by the size and ripeness of the fruit. If immersed too long, the pulp is either over-

cooked, or is left too dry and woody. If taken out too soon, the juices left in the pulp prevent perfect absorption of the sugar afterwards, and by eventually causing fermentation, destroy the value of the product. A skilful workman can tell by the colour and appearance of the pulp when it is properly 'blanched.' For the different grades of fruits, sugar-sirups of different degrees of density are required: the softer the fruit, the stronger the sirup required for its preservation. The sirup having been prepared by dissolving the sugar in pure water, the fruit is immersed in it and left at rest for a certain period in large earthenware pans, glazed inside. The sirup penetrates the pulp, and gradually withdraws and replaces the remaining fruity juice, which, as it exudes and mingles with the transparent liquid, produces a certain filmy or clouded appearance, which marks the commencement of fermentation. When this has reached a certain stage, the vessel containing the sirup and fruit is placed over the fire and heated to two hundred and twelve degrees, which corrects the fermentation. If the sirup is of proper density, the process of impregnating the fruit with sugar will be complete in about six weeks, during which period it is sometimes necessary to perform the heating process three times. The fruit now goes through one of two finishing processes according as it is to be 'glazed' or 'crystallised.' Some manufacturers are said to quicken the crystallisation of fruit by the use of a powerful antiseptic called salicylic acid; but although time, labour, and sugar are thereby saved, Mr Mason believes it is at the expense of quality in the finished product.

THE ANCIENT BOAT AT BRIGG.

A notice will be found in [No. 126](#) of the *Journal* referring to the discovery, at Brigg in Lincolnshire, during the excavations for a new gasholder, of a curious and ancient boat cut out of a solid piece of oak, and measuring forty-eight feet in length, fifty-two inches in width, and thirty-three inches in depth. The vessel is in a fine state of preservation, and it is to be hoped that proper means will be provided by the authorities for preserving this interesting relic. The last news that we have of it, however, is that it has 'got into Chancery.' A curious dispute seems to have arisen as to the ownership of this relic; and probably, when the case comes to be argued before the Court, some interesting legal points will be raised by the gentlemen of the 'long robe' as to the main question at issue. Whatever may be the result, one

thing is certain, that so rare a prehistoric relic as this should be preserved to the nation as public property, on the spot, or in the town near to where it was found, as an object of peculiar local interest. It would be a mistake to remove it to London, as has been suggested; but to exhibit it for money is neither fair nor proper, and the public will probably watch the proceedings before the High Court of Chancery with interest. Boats found buried in the earth and dating from remote antiquity are very rare in this country, although several have been discovered of late years in Norway and Denmark, they having been the tomb or grave of the original commander, one of the brave and lawless vikings who roamed the seas and ravaged the neighbouring coasts of Europe in search of conquest and plunder, and when at last his restless life had closed, made his beloved ship at once his monument and sepulchre.

RELICS OF ANCIENT CARTHAGE—MOSAICS.

Not long ago, some highly artistic relics of ancient Carthage were disposed of at an auction in London. Two of the finest of these are mosaics in splendid preservation, each about three feet square. The one represents a woman robed and wearing a crown of flowers, with a naked youth sitting beside her; and the other a youth carrying on his shoulders an eagle. These have been called 'Peace' and 'War;' but there seems to be no authority for this. Both works are evidently early Carthaginian, and must have belonged to a period when Carthage held a high position as a nursery of art, especially in the beautiful art of mosaic-work, of which ancient Greece has left no trace, whilst the mosaics of Rome are of a much later date. It will be remembered that Carthage was celebrated for her beautiful coloured marbles, and for the wonderful skill of her artists and workmen, which were known throughout the civilised world, for Carthage was a large city one hundred and forty years before the foundations of Rome were laid. It is possible, therefore, that the peculiar art of working in mosaic may have been originated in Carthage, and may have found its way to Rome, where it might have been practised by Roman, or even Carthaginian artists. But, as a rule, the Roman work is very inferior to the Carthaginian. These specimens were, with many others, collected by Count d'Hérissou from recent excavations made in a garden at Danar-el-Sciath, near Tunis, and situated in the midst of the ruins of ancient Carthage. Of the authenticity of these relics

there can be, therefore, no possible doubt, as they were brought direct from the site of the city itself. The two referred to, together with several other interesting specimens, were purchased by Mr Edwin Long, R.A.

Whilst on the subject of mosaics, we may mention that a valuable discovery has just been made at Chiusi in Italy. Whilst some workmen were digging out a watercourse at the foot of a hill near Monte Venere, they came upon a mosaic pavement about nine feet by six feet in size. The centre represents a double hunting scene: in the top row are three stags pursued by a hunter with a spear; below is a boar followed by two hunters, carrying each an axe and lance. The whole work is in perfect preservation, well and carefully executed with much fire and spirit, and is interesting as being the first piece of mosaic pavement that has been discovered in Chiusi or its neighbourhood.

SWEETHEART, FAREWELL.

BENEATH the whispering trees we lingered late,
Hand clasped in hand, my dearest love and I,
And he spake words I never can forget,
Of tender trust and love, until I die;
And with his eyes what lips would fail to tell
He spoke, what time he said: 'Sweetheart, farewell.'

With sweet caress he clasped me to his breast,
And looked upon me as with angel's eyes,
And kissed my brow, and kissed my lips, and kissed
The tears away that now began to rise;
And ever the same tale of love would tell,
What time he sadly spoke: 'Sweetheart, farewell.'

And so he went away, and I am weary
Of nature's smiles—my heart is full of strife—
The long, long days without him are so dreary,
And all the bright has faded out of life.
'Come back, my love, the old sweet tale to tell,
But nevermore to say: "Sweetheart, farewell."'

WILLIAM COWAN.

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